

Background

Within the next 13 years the elderly population is predicted to double. Ranked at number six in the leading causes of death, Alzheimer’s is a progressive degenerative disease with no cure. Accounting for 60-80 percent of all cases, Alzheimer’s disease (AD) is the most prevalent form of dementia. The cognitive deficits observed in AD are directly related to the loss of synaptic function and reduction in the quantity and density of dendritic spines. The structure of each dendritic spine is speculated to reflect its function (Dorostkar, Zou, Blazquez-Llorca, Herms. 2015). Additionally, two thirds of the individuals affected by AD are females. Although there is currently no cure for AD, treatments exists to slow dementia symptoms and increase overall quality of life. The race for a cure continues as the prevalence of AD continues to rise.

Hypothesis and Aims

We hypothesize that if chronic inflammation is induced at 12 weeks of age for six weeks, then by four months of age it is predicted these mice will have fewer dendritic processes and dendritic spines throughout the hippocampus and cortex.

Aims:

- Determine how neuroinflammation affects neuronal ramification and dendritic spines in regards to gender and genotype.
- Demonstrate the role of neuroinflammation as a potential gender-related risk factor in AD progression.

Methods and Animals

Weekly intraperitoneal LPS or vehicle injections of 1 mg/kg for six weeks.

Confocal Microscopy

The confocal microscope uses a spatial pinhole to block interfering light and increase resolution while compiling a 3D reconstruction of the dendrites present in the tissue sample.

Six neurons per tissue slice (three – cortex; three – hippocampus), 5 slices per animals.

Parameters: 1024 ¼ (speed) ~150µm 2 steps pixel size of 0.3 20X HV (exposure) = 114

Neurolucida

Program that quantifies neuronal morphology by providing the length, width and volume of the each neuron’s dendritic spines.

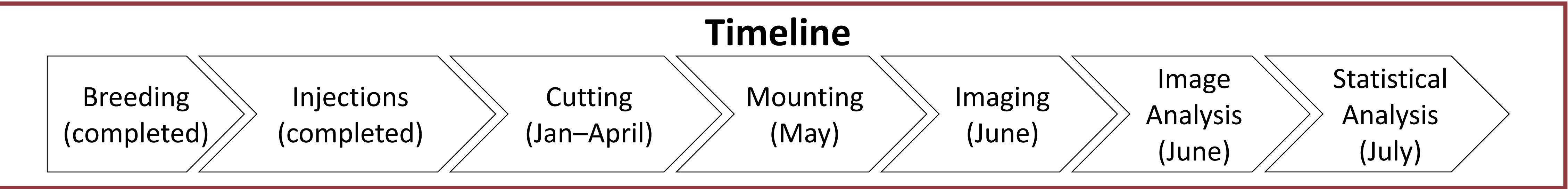
Animals

We are using a new knock-in mouse model of Alzheimer’s disease containing the Swedish, Beyreuther/Iberian and Arctic mutations, *APP^{N-L-GF}* (Saito, T. et al. 2014). Groups: n = 5 to 7 per group

	Heterozygous	Homozygous	WT
♂	LPS/Vehicle	LPS/Vehicle	LPS/Vehicle
♀	LPS/Vehicle	LPS/Vehicle	LPS/Vehicle

Perspective

This study’s goal is to elucidate if inflammation is a factor in AD progression and thus can be targeted to prevent or delay AD.



Saito, T., Matsuba, Y., Mihira, N., Takano, J., Nilsson, P., Itohara, S.,...Saito, T. C. (2014). Single App Knock-in mouse models of Alzheimer’s disease. Nature Neuroscience, 17(5), 661-663.

Dorostkar, M. M., Zou, C., Blazquez-lorca, L., & Herms, J. (2015). Analyzing dendritic spine pathology in alzheimer’s disease: Problems and opportunities. *Acta Neuropathologica*, 130(1), 1-19.

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